Anyu Pan

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RESEARCH INTERESTS

Agentic AI; AI for Healthcare; Computer Vision; Computational Drug Discovery

EDUCATION

University of Pennsylvania, USA 2024-2027

- M.S. in Computer and Information Science
- M.S. in Bioengineering

University of Dundee, UK 2020-2024

Joint Undergraduate Programme with Northeastern University, China

- B.Eng. in Biomedical Engineering (First Class Honours)

RESEARCH EXPERIENCES

Shen Lab | University of Pennsylvania 06/2026-Present

Graduate Student Researcher | Penn Engineering (CIS) & Penn Medicine | PI: Li Shen, Ph.D.

Alzheimer's Disease Drug Discovery 07/2026-Present

- Built an **AGORA-centered target database** integrating BindingDB, ChEMBL, UniProt, RCSB PDB, Drugs@FDA, and ClinicalTrials.gov using **DuckDB**; linked **967 targets, 452,249 compounds, and 1,430,260 measurements**
- Curated a PIK3CA potency dataset and fine-tuned the **Boltz-2 affinity head**; improved Pearson r from **0.248 to 0.560** and reduced MAE from **0.900 to 0.706** on **469 held-out compounds**
- Developed and deployed AGORA Target & Drug Evidence Explorer on **Azure**, providing a public platform for AD target assessment, drug-target evidence queries, curated data exports, and access to fine-tuned models and evaluation results

Mouse Scratching Classification with Video Transformers 06/2026-Present

- Fine-tuned **Video Swin Transformer** and adapted **InternVideo-Next** with low-rank adaptation (LoRA) for **Scratch-AID+**, a framework for automated mouse scratching detection
- Benchmarked on held-out videos from four external labs under realistic sparse-event evaluation; InternVideo-Next achieved **81.19% mean scratching-class F1**, exceeding Scratch-AID by **14.88 percentage points**, averaged over six preprocessing configurations

Advancing Glioma Classification with Swin Transformers | Honours Project 09/2023-05/2024

University of Dundee & University of Cambridge

- Integrated **Swin Transformer blocks** into a WSI-based glioma classification framework, improving accuracy to 78.03% (from 73.11% with a standard ViT)
- Systematically evaluated multiple **optimizers** (SGD, Momentum, AdaGrad, RMSprop, Adam, and Lion) to identify optimal performance-efficiency trade-offs
- Reduced **GPU memory usage by >60%**, enabling larger batch sizes and more efficient GPU training

PUBLICATIONS

Yang, S.*, **Pan, A.***, Shin, H.*, Chen, H.*, Zhang, Z., Shad, A., Huang, W., Liu, Y., Liu, W., Luo, H., Xu, C., Chen, J., Xu, M., Luo, W., Yu, H., & Shen, L. (2026). Towards Robust Automated Mouse Scratching Behavior Detection for Itch Research. *AI4BioSig 2026 (IEEE BIBM 2026 Workshop), accepted.* [Paper]

* Equal contribution (co-first authors).

Pan, A., Zhai, T., Lee, J., Zheng, C., Kim, D., & Shen, L. (2026). Integrating Multisource Target Evidence with Machine-Learning Potency Prediction for Alzheimer's Disease Drug Discovery. *Submitted to the AMIA 2027 Amplify Informatics Summit* (regular paper). [Paper]

Qiu, Z., **Pan, A.**, & Qi, L. (2023). Incorporating dynamic attention gating mechanism and pre-trained embedding for Chinese clinical named entity recognition. *Procedia Computer Science, 226*, 113-119.

SELECTED RESEARCH & AI PROJECTS

WeatherWise Trip Concierge | Google/Kaggle AI Agents Capstone 06/2026-07/2026

- Built a **Google ADK travel-planning agent** with **MCP tools** to retrieve forecasts, score daily weather risks, and recommend nearby cities
- Implemented **rule-based risk scoring** and Pydantic input validation; tested prompt-injection and missing-forecast cases
- Created **AGENTS.md**, project context documentation, and reusable **agent skills**; deployed the web app on **Google Cloud Run** using Docker

Brain-Computer Interface: Decoding Finger Flexion from ECoG | Member 03/2025-05/2025

- Developed a **convolutional encoder–decoder** to predict five-finger flexion from ECoG (BCI Competition IV dataset)
- Built signal preprocessing with bandpass/notch filters, **Morlet wavelets**, downsampling, and lag correction
- Achieved mean Pearson correlation of 0.624 on the held-out test set using a combined **MSE + cosine loss**

Biomedical Image Analysis | Member 11/2024-12/2024

- Preprocessed MRI scans and tuned **N4 bias field correction** to improve segmentation inputs
- Quality-checked 3D training patches, labeled MRI in **ITK-SNAP**, and merged team labels into ground-truth masks
- Trained and evaluated an **nnU-Net** segmentation pipeline, refining hyperparameters to improve accuracy

The 8th National College Student Biomedical Engineering Innovation Design Competition | Core Member 05/2023-06/2023

- Integrated **YOLOv5 and Transformer models** for rapid colorectal polyp detection and diagnosis
- Reduced the missed detection rate from **22% to 1%** in simulations

SELECTED CS/ML PROJECTS

Internet and Web Systems: Distributed Search Engine | Member 11/2025-12/2025

- Hardened a crawler with HTTP/socket timeouts and robots handling; crawled **100K+ pages in ~9 hours**
- Optimized distributed **PageRank** using per-partition KVS writes; fixed persistence and Flame partitioning, reducing client churn
- Added autocomplete and normalized URLs, titles, and snippets; deployed on **AWS EC2** and benchmarked instance types

Big Data Analytics: Spotify Popularity Prediction | Member 10/2024-12/2024

- Preprocessed **114,000 Spotify records** and used Plotly for interactive exploration of popularity-related features
- Trained and tuned **Random Forest and XGBoost** regression models for song popularity prediction

Applied Machine Learning: Knock2Know (Audio Classification) | Member 09/2024-12/2024

- Collected and preprocessed audio across recording devices, times of day, and knocking techniques to improve robustness
- Extracted **OpenAI Whisper** features and trained a shallow CNN with augmentation and early stopping; achieved **0.71 accuracy** on unseen samples

The 16th Computer Design Competition for Chinese College Students | Member 04/2023-08/2023

- Developed a **Flutter** game about traditional Chinese medicine; led debugging, documentation, and regression testing

Mathematical Contest in Modeling: Breaking the Wordle | Core Member 02/2023

- Modeled Wordle trends with **Prophet** and built predictive models using multi-objective regression and **LASSO**.

TEACHING EXPERIENCE

Developmental Engineering of Tissues | Teaching Assistant | Penn Engineering 08/2024-12/2024

- Graded code and reports for a neural crest migration model, covering Voronoi adjacency, border-cell detection, force balances, and numerical integration; provided detailed feedback
- Held weekly office hours on debugging and model design; graded quizzes to assess preparation and progress

AWARDS & HONORS

- **2023 ASA Data Insights Award (top 0.1%)**, Mathematical Contest in Modeling, International Level 11/2023
- **Outstanding Winner (top 0.2%), COMAP Scholarship Award & ASA Award**, Mathematical Contest in Modeling, International Level 05/2023
- **First Prize (top 4%)**, 16th Computer Design Competition for Chinese College Students, National Level 08/2023

SKILLS

Programming Languages: Java, Python, C/C++, SQL, R, MATLAB, HTML, JavaScript, CSS, Dart

Machine Learning: PyTorch, scikit-learn, XGBoost, ViT, Video Swin Transformer, InternVideo-Next, LoRA, Boltz-2, YOLO, nnU-Net, NumPy, SciPy, pandas

Agent Development: Google ADK, MCP, Agent Skills, Docker, Google Cloud Run

Research Infrastructure: RDKit, DuckDB, Azure, Slurm

Others: AWS, Flutter, Qt, Git, LaTeX, Arduino IDE, InVesalius, ITK-SNAP, SolidWorks, Proteus

ACADEMIC EXCHANGE

Osaka University, Japan - Exchange Program in Biotechnology 05/2021-08/2021